ZERO DIVISORS OF THE INTEGRAL SEDENIONS AND THE COMPACT G_2

PART IV OF THE OCTONION PROGRAM RECORD

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Abstract. We study the zero divisors of the integral sedenions, the Cayley–Dickson double of Coxeter's order of integral octonions, and their interaction with the arithmetic of the compact group G_2. Among the 57,600 sedenions whose two octonion halves are units, exactly 7,560 are zero divisors, namely those whose halves are imaginary and orthogonal; they fall into two $G_2(\mathbb{Z})$-orbits of sizes 6,048 and 1,512, distinguished by the type (Lipschitz or Hurwitz) of the quaternion subalgebra they generate, and the 84,672 ordered pairs (x, y) with $xy = 0$ form seven free $G_2(\mathbb{Z})$-orbits: the integral shadow of Moreno's theorem that the real unit zero-divisor pairs form the Lie group G_2. Over $\mathbb{R}$ the zero-divisor manifold is $G_2/\mathrm{Sp}(1)$ and its Hecke algebra $\mathcal{H}(G_2, \mathrm{Sp}(1))$ is computed in low weights. The conic of a quaternion subalgebra modulo p meets the maximal-parahoric class set of the compact G_2 at p in a pattern governed by the octahedral group S_4: for $p \in \{3, 5, 7, 11\}$ each conic lies in a single class determined by the type of the zero divisor, and for $p \geq 13$ it splits into S_4-orbits, with class stabilisers $192/m$ and $48/m$; this is proved for all $p \geq 5$. Zero-divisor periods of algebraic modular forms on G_2 at level 3 are defined; they are the D^1-periods of Gan and Savin's exceptional see-saw, and the period along the conic separates Iwahori-new forms from the old ones (through the degeneracy map), which are identified exactly as Gan–Gurevich-type lifts of the level-3 newforms of weights 6 and 10 with the classical form on D^1. Three constructions of a sedenion spectral geometry in the sense of Farnsworth and Boyle are shown to admit no finite Dirac operator beyond a scalar mass, and the Euler product of the norm-counting zeta function is shown to survive exactly up to the sedenions along the Cayley–Dickson ladder. All computations and code are deposited in the program's repository.

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1. Introduction and summary

Parts I and II of this record [1, 2] concern class sets and Hecke modules of the compact groups G_2 and F_4 built from Coxeter's integral octonions $\mathbb{O}_\mathbb{Z}$ (the E_8 lattice). The present part goes one rung further on the Cayley–Dickson ladder, to the sedenions $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$, and asks what the first failure of division has to do with the arithmetic developed so far. The answer is more structured than expected.

(a) The integral sedenions $\mathbb{S}_\mathbb{Z} = \mathbb{O}_\mathbb{Z} \oplus \mathbb{O}_\mathbb{Z}\ell$ form the lattice $E_8 \oplus E_8$. Their zero divisors with unit halves are *exactly* the elements $a + b\ell$ with a, b imaginary orthogonal units (Theorem 2.1); there are 7,560 of them, each with a 4-dimensional annihilator, and they form two $G_2(\mathbb{Z})$-orbits separated by the type of the quaternion subalgebra $\langle 1, a, b, ab\rangle$. The ordered zero-divisor pairs form seven free $G_2(\mathbb{Z})$-orbits (Theorem 2.2).

(b) Over $\mathbb{R}$ the unit zero divisors form $G_2/\mathrm{Sp}(1)$ and Moreno's G_2 is the sphere bundle of their annihilators. The Hecke algebra $\mathcal{H}(G_2, \mathrm{Sp}(1))$, a non-commutative algebra with continuous families of idempotents, has blocks of sizes $1, 3, 3, 6, 8, 10, 6$ on the representations of dimensions $1, 7, 14, 27, 64, 77, 77'$ (Section 3).

(c) The conic law (Theorem 4.4): the isotropic lines of $\mathrm{Im}\,\mathbb{H}_z \otimes \mathbb{F}_p$ form a conic $\cong \mathbb{P}^1(\mathbb{F}_p)$, and its intersection with the maximal-parahoric class set of the compact G_2 at p is governed by the octahedral group S_4. The two zero-divisor types index two classes for $p = 3, 5, 7, 11$ and the conics split into cube-orbits for $p \geq 13$.

(d) Zero-divisor periods of algebraic modular forms (Section 5) are the D^1-periods of the Gan–Savin see-saw [6]; the period along the conic vanishes exactly on the Iwahori-new forms of weight $2\omega_1$ at level 3, while their periods on other orbits do not.

(e) Negatives (Section 7): in the Farnsworth–Boyle framework of non-associative spectral geometry [8, 9] the sedenions admit, in all constructions tried, no finite Dirac operator beyond a scalar mass; the sedenion analogue of the Albert algebra is not Jordan; and the Euler product of the norm-counting zeta function dies one doubling past the sedenions, where Ramanujan's Δ first enters.

Everything in (a)–(d) was computed exactly (integer arithmetic, finite groups, or linear algebra at machine precision with independent checks); the conic law is proved for all $p \geq 5$ by a structural reduction to a finite computation (Section 4).

2. The integral sedenions and their zero divisors

Let $\mathbb{O}_\mathbb{Z}$ be Coxeter's maximal order of integral octonions, with norm N, trace tr and the 240 units. The Cayley–Dickson double $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$ has product $(a, b)(c, d) = (ac - \bar{d}b,\ da + b\bar{c})$, conjugation $\overline{(a, b)} = (\bar{a}, -b)$ and norm $N(a, b) = N(a) + N(b)$; we take $\mathbb{S}_\mathbb{Z} = \mathbb{O}_\mathbb{Z} \oplus \mathbb{O}_\mathbb{Z}\ell$. On 200 random triples the laws behave as classically known: $\mathbb{S}$ is flexible and power-associative; it is not commutative, associative or alternative, and $N(xy)/(N(x)N(y))$ ranged over $[0.709, 1.316]$.

Theorem 2.1. *Let $x = (a, b) \in \mathbb{S}_\mathbb{Z}$ with $N(a) = N(b) = 1$. Then x is a zero divisor if and only if a and b are both imaginary ($\mathrm{tr} = 0$) and orthogonal. There are 7,560 such elements; each has a 4-dimensional annihilator $\{y : xy = 0\}$. No unit of $\mathbb{S}_\mathbb{Z}$ (norm 1) is a zero divisor.*

Proof. Exhaustive computation over all $240^2 = 57{,}600$ pairs (a, b) of units: the left-multiplication matrix L_x is singular exactly for the 7,560 pairs with $\operatorname{tr} a = \operatorname{tr} b = 0$ and $\langle a, b\rangle = 0$ (the other seven combinations of the three conditions never give a zero divisor), and its kernel has dimension 4 in every case. A unit has one half zero and all 230,400 products of units have norm 1. □

The real version (imaginary, orthogonal, equal norm) is due to Moreno and to Khalil–Yiu; the count and the dimension 4 reproduce Moreno's dimension count $11 + 3 = 14 = \dim G_2$ for the set of unit zero-divisor pairs [3].

Theorem 2.2. *The group $G_2(\mathbb{Z}) = \operatorname{Aut}(\mathbb{O}_\mathbb{Z})$ (order 12,096) acts on the 7,560 zero divisors with two orbits, of sizes 6,048 and 1,512, with stabilisers of orders 2 and 8. The quaternion subalgebra $\mathbb{H}_z = \langle 1, a, b, ab\rangle \cap \mathbb{O}_\mathbb{Z}$ contains 8 units (Lipschitz type) on the large orbit and 24 (Hurwitz type) on the small one, and these are exactly the numbers of zero divisors y in the annihilator of x among the 7,560. The $84{,}672 = 7 \cdot 12{,}096$ ordered pairs (x, y) of these zero divisors with $xy = 0$ form seven free $G_2(\mathbb{Z})$-orbits.*

Proof. Computation with the 12,096 automorphisms in the Coxeter basis (each verified to be a sedenion automorphism acting diagonally on the two halves); the orbit and stabiliser counts are exact. The map $(a, b) \mapsto (a, -b)$ is an automorphism of $\mathbb{S}$; the half-swap preserves the zero-divisor set but is not an automorphism. □

Remark 2.3. Moreno's theorem says the real unit zero-divisor pairs form a space homeomorphic to G_2; Theorem 2.2 is its integral shadow: seven copies of $G_2(\mathbb{Z})$, acted on freely.

3. The zero-divisor manifold and its Hecke algebra

Over $\mathbb{R}$, a unit zero divisor is a pair (a, b) of orthonormal imaginary octonions. G_2 acts transitively on such pairs, so the zero-divisor manifold is $ZD = G_2/K$ with K the stabiliser of one pair.

Proposition 3.1. *K is 3-dimensional with Lie algebra $\mathfrak{su}(2)$; it fixes $\mathbb{H}_z$ pointwise and acts on the 4-dimensional annihilator of $z = (a, b)$ as $\mathrm{Sp}(1)$ acts on $\mathbb{H}$ (the commutant of K on the annihilator is 4-dimensional). Hence $ZD = G_2/\mathrm{Sp}(1)$ is the Stiefel manifold of orthonormal pairs in $\mathbb{R}^7$, of dimension 11, and Moreno's G_2 is the unit-sphere bundle of the annihilators over it.*

Proof. Computed from the 14 derivations of $\mathbb{O}$ in an orthonormal basis: the derivations killing a and b form a 3-dimensional subalgebra, which kills $\langle 1, a, b, ab\rangle$ and preserves the annihilator, on which its commutant has dimension 4. □

In Gan–Savin's notation [6], $K = D^1$ is the group of norm-one elements of the quaternion subalgebra $D = \mathbb{H}_z$, acting by $g \cdot (a, b) = (a, b\bar{g})$; it is the pointwise stabiliser of D in $\operatorname{Aut}(\mathbb{O})$.

The Hecke algebra of the homogeneous space is $\mathcal{H}(G_2, K) = \bigoplus_\pi M_{m_\pi}(\mathbb{C})$ with $m_\pi = \dim \pi^K$ (Frobenius reciprocity: m_π is the multiplicity of π in $L^2(ZD)$). Decomposing $7 \otimes 7$, $7 \otimes 14$, $\mathrm{Sym}^3 7$ and $\mathrm{Sym}^2 14$ by the Casimir and counting K-fixed vectors:

π	1	7	14	27	64	77	77′
$m_\pi = \dim \pi^K$	1	3	3	6	8	10	6

The 7 gives 3 because K fixes exactly $\operatorname{Im}\mathbb{H}_z \subset \operatorname{Im}\mathbb{O}$; the 14 gives 3 because the centraliser of K in $\mathfrak{g}_2$ is the other $\mathfrak{su}(2)$ of $\mathfrak{so}(4) \subset \mathfrak{g}_2$. Since (G_2, K) is not a Gelfand pair, the idempotents of each block form continuous families (Grassmannians). The representation theory here is classical; the identification of ZD with the sedenion zero divisors is what makes it ours. Every weight in the Path A table of compact G_2 forms (Part II) has $m_\pi > 0$, which is what makes the periods of Section 5 possible.

4. Quaternion subalgebras and the conic law

4.1. **Quaternion subalgebras of the Coxeter order.** Every zero divisor $z = (a, b)$ determines $\mathbb{H}_z = \langle 1, a, b, ab\rangle \cap \mathbb{O}_{\mathbb{Z}}$, a quaternion order of the definite quaternion algebra it spans. Counting ordered orthogonal imaginary unit pairs per subalgebra ($6 \cdot 4 = 24$ in both cases) gives:

Proposition 4.1. *$\mathbb{O}_{\mathbb{Z}}$ contains 63 quaternion subalgebras of Hurwitz type (24 units) and 252 of Lipschitz type (8 units). Their stabilisers in $G_2(\mathbb{Z})$ have orders 192 and 48; the pointwise stabilisers have orders 8 and 2 (the stabilisers of the zero divisors of Theorem 2.2), and in both cases the quotient acting faithfully on $\operatorname{Im}\mathbb{H}_z$ is the octahedral group S_4.*

Proof. $1512/24 = 63$, $6048/24 = 252$; $12096/63 = 192$, $12096/252 = 48$; the pointwise stabilisers were computed directly, and $192/8 = 48/2 = 24 = |S_4|$, with S_4 the full rotation group of the cube $\{\pm i, \pm j, \pm k\}$, realised in both cases by the computation of Section 5 (orbit sizes $6, 8, 12$ below). □

Proposition 4.2. *The imaginary integral units are the 126 roots of the E_7 lattice $\operatorname{Im}\mathbb{O}_{\mathbb{Z}}$, and the 7,560 zero divisors of Theorem 2.1 are exactly the ordered pairs of orthogonal E_7 roots (60 roots are orthogonal to any given root, a D_6). The zero-divisor mass $7560/12096 = 5/8$ is thus a Siegel representation number of the binary form $2I_2$ by E_7 divided by $|G_2(\mathbb{Z})|$, and the zero divisors of each norm are governed by Siegel's local–global formula for representations by E_7, even though their counting series has no Euler product of its own.*

Proposition 4.3. *For a zero-divisor pair $x = (a, b)$, $y = (c, d)$ with $xy = 0$ one has $d = \pm(ac)b$, so the pair is the triple (a, b, c) with c a unit perpendicular to $\mathbb{H}_x$. The seven free $G_2(\mathbb{Z})$-orbits of Theorem 2.2 are separated by the types (Hurwitz 24 or Lipschitz 8) of the five quaternion subalgebras $\langle a, b\rangle, \langle c, d\rangle, \langle a, c\rangle, \langle b, c\rangle, \langle ab, c\rangle$: the seven realised type-patterns are $(8, 8, 8, 24, 8)$, $(8, 8, 24, 8, 8)$, $(8, 24, 8, 8, 24)$, $(8, 24, 24, 24, 24)$ for Lipschitz x and $(24, 8, 8, 24, 8)$, $(24, 8, 24, 8, 8)$, $(24, 24, 8, 8, 24)$ for Hurwitz x, with equal counts within each type as the freeness requires.*

4.2. **The conic.** Fix an odd prime p. The space $\operatorname{Im}\mathbb{O} \otimes \mathbb{F}_p$ is a nondegenerate 7-dimensional quadratic space; its isotropic lines number $(p^6-1)/(p-1)$ and form the maximal-parahoric level-p structures of the compact G_2, so that the class set is $X_p = G_2(\mathbb{Z})\backslash\{\text{isotropic lines}\}$ (Part I). For a zero divisor z, $\operatorname{Im}\mathbb{H}_z \otimes \mathbb{F}_p$ is a nondegenerate ternary space, so its isotropic lines form a conic $C_z \cong \mathbb{P}^1(\mathbb{F}_p)$ with $p+1$ points, on which $\operatorname{Stab}(\mathbb{H}_z)$ acts through $S_4 \subset \mathrm{PGL}_2(\mathbb{F}_p)$.

Theorem 4.4 (Conic law)**.** *Let $p \geq 5$ and z a zero divisor. The $G_2(\mathbb{Z})$-classes of the points of C_z are exactly the S_4-orbits on $\mathbb{P}^1(\mathbb{F}_p)$, whose sizes lie in $\{6, 8, 12, 24\}$ (cube faces, vertices, edges, and generic), and the stabiliser in $G_2(\mathbb{Z})$ of a conic point in an orbit of size m has order $192/m$ (Hurwitz type) or $48/m$ (Lipschitz type). The number of classes a conic meets is the number of S_4-orbits on $\mathbb{P}^1(\mathbb{F}_p)$, which by Burnside's lemma is*

$$\frac{1}{24}\Big((p{+}1){+}3f_{-1}{+}6f_{-2}{+}8f_3{+}6f_4\Big), \qquad f_{-1} = 2 \text{ if } \left(\tfrac{-1}{p}\right) = 1,\ f_{-2} = 2 \text{ if } \left(\tfrac{-2}{p}\right) = 1,\ f_3 = 2 \text{ if } p \equiv 1\ (3),\ f_4 = 2 \text{ if}$$

and 0 otherwise (the face, edge and vertex rotations of the cube have 2 or 0 fixed points on the conic according to these conditions). Equivalently, the conic carries a 6-point orbit (stabilisers 8 and 32) exactly when $p \equiv 1\ (4)$, an 8-point orbit (6 and 24) exactly when $p \equiv 1\ (3)$, a 12-point orbit (4 and 16) exactly when $(\frac{-2}{p}) = 1$, and all remaining points lie in 24-point orbits (2 and 8); this was verified directly for all $p \leq 41$. The orbit structure is thus Eisenstein data of level 24: the number of classes a conic meets is $\frac{p+1}{24} + \frac{3}{4}[p \equiv 1\,(4)] + \frac{1}{2}[(\frac{-2}{p}) = 1] + \frac{2}{3}[p \equiv 1\,(3)]$, and its Dirichlet series over p is a combination of $\zeta(s-1)$, $\zeta(s)$ and the L-functions of the characters of conductors 3, 4 and 8, while the cuspidal content of the zero divisors lies entirely in the periods

of Section 5. In particular each conic lies in a single class exactly for $p \in \{5, 7, 11, 23\}$ *(*$p = 23$ *is the regular orbit,* 24 *points with stabilisers* 2 *and* 8*), and for* $p \geq 29$ *never; for* $p = 3$ *the conic (*4 *points) also lies in a single class, with the exceptional stabilisers* 432 *and* 36*. The count depends only on* p *modulo* 24*.*

Lemma 4.5. *For* $p \geq 5$*, an isotropic line lying on the conic of a quaternion subalgebra of given type lies on no other conic of that type.*

Proof. Let $\mathbb{H}_1 \neq \mathbb{H}_2$ be suborders of the same type, $M_i = \operatorname{Im} \mathbb{H}_i$ (rank 3). Since $\mathbb{H}_i = B_i \cap \mathbb{O}_\mathbb{Z}$ for the $\mathbb{Q}$-subalgebra B_i it spans, $B_1 \neq B_2$, and two distinct quaternion subalgebras of $\mathbb{O}$ meet in a subalgebra of dimension ≤ 2 (a 3-dimensional subalgebra of a quaternion algebra does not exist); hence $M_1 \cap M_2$ has rank ≤ 1. The conics C_1, C_2 modulo p share a point only if $\bar{M}_1 \cap \bar{M}_2 \subset \operatorname{Im} \mathbb{O} \otimes \mathbb{F}_p$ contains a nonzero isotropic vector. Now $\dim_{\mathbb{F}_p}(\bar{M}_1 \cap \bar{M}_2) = 6 - \operatorname{rank}_p(M_1 + M_2)$, and $\operatorname{rank}_p(M_1 + M_2) = \operatorname{rank}(M_1 + M_2)$ unless p divides the index of $M_1 + M_2$ in its saturation. So either (i) p divides that index, or (ii) $\bar{M}_1 \cap \bar{M}_2$ is the reduction of $M_1 \cap M_2$, which is 0 or $\mathbb{Z}v$ with v a common imaginary vector, isotropic mod p only if $p \mid N(v)$.

Computation over all $252 \cdot 251/2 = 31{,}626$ Lipschitz pairs and $63 \cdot 62/2 = 1{,}953$ Hurwitz pairs (code `g2table/lemma43.py`): the saturation indices take the values $1, 2, 3, 4, 5, 6, 8, 10$ (Lipschitz) and $1, 2, 3, 4, 5, 6, 7, 8$ (Hurwitz), so their prime factors lie in $\{2, 3, 5, 7\}$; the pairs with $\operatorname{rank}(M_1 + M_2) = 5$ (6,426 Lipschitz, 1,197 Hurwitz) have common directions of norm 1 or 2 (Lipschitz) and 1 or 3 (Hurwitz). Hence for $p \geq 11$ case (i) never occurs and case (ii) gives no isotropic vector: the conics are disjoint. For $p = 5$ and $p = 7$, case (i) occurs for $15 + 75$ pairs at $p = 5$ and 23 Hurwitz pairs at $p = 7$; for each of these the intersection $\bar{M}_1 \cap \bar{M}_2$ was computed over $\mathbb{F}_p$ and found to contain no isotropic vector, while case (ii) is excluded as before since $p \nmid N(v) \in \{1, 2, 3\}$. This proves the lemma for all $p \geq 5$. (For $p = 3$ it fails: $3 \mid N(v)$ for the Hurwitz pairs with a common direction of norm 3, and the conics do share points, which is why the stabilisers $432, 36$ at $p = 3$ exceed $192/4$ and $48/4$.) □

Proof of Theorem 4.4 from Lemma 4.5. Let $\ell \in C_z$ and $g \in G_2(\mathbb{Z})$ with $g\ell \in C_z$. Then $g\ell$ lies on the conics of $\mathbb{H}_z$ and of $g\mathbb{H}_z$, both of the type of z; by the lemma $g\mathbb{H}_z = \mathbb{H}_z$, so $g \in \operatorname{Stab}(\mathbb{H}_z)$. Hence the $G_2(\mathbb{Z})$-orbit of ℓ meets C_z exactly in its $\operatorname{Stab}(\mathbb{H}_z)$-orbit, which is its S_4-orbit since the pointwise stabiliser acts trivially on C_z, and the stabiliser of ℓ in $G_2(\mathbb{Z})$ equals its stabiliser in $\operatorname{Stab}(\mathbb{H}_z)$, of order $|\operatorname{Stab}(\mathbb{H}_z)|/m$. The orbit sizes of $S_4 \subset \mathrm{PGL}_2(\mathbb{F}_p)$ on $\mathbb{P}^1(\mathbb{F}_p)$ are $24/|\text{point stabiliser}|$ with point stabilisers the cyclic subgroups of orders $4, 3, 2, 1$, giving $6, 8, 12, 24$; S_4 is transitive on $\mathbb{P}^1(\mathbb{F}_p)$ iff $p + 1 \in \{6, 8, 12\}$ (the case 4 being $p = 3$, where the lemma fails and the stabilisers are larger). □

Computation 4.6. Independently of the proof, exact orbit computations for $p = 3, 5, 7, 11, 13, 17, 19$ (four to forty zero divisors of each type per prime) give the class structure of the conic points shown in Table 1; the stabiliser identity $|\operatorname{Stab}(\ell)| = |\operatorname{Stab}(\mathbb{H}_z)|/m$ holds in every case with $p \geq 5$, as the theorem predicts. The classes touched were also matched against the full class sets X_5 (3 classes, stabilisers $32, 8, 6$: the 6-class is touched by no zero divisor) and X_7 (8 classes).

Remark 4.7. The incidence counts are now explained: $63(p + 1)$ Hurwitz and $252(p + 1)$ Lipschitz incidences fall on classes of sizes $12096\,m/192$ and $12096\,m/48$, one conic through each point.

5. Zero-divisor periods

5.1. **Definition.** Let f be an algebraic modular form of weight π for the compact G_2 at a parahoric level at $p = 3$: a function on the level-3 structures x (isotropic lines, planes or flags

p	$p+1$	Lipschitz: (points, stabiliser)	Hurwitz: (points, stabiliser)
3	4	$(4, 36)$	$(4, 432)$
5	6	$(6, 8)$	$(6, 32)$
7	8	$(8, 6)$	$(8, 24)$
11	12	$(12, 4)$	$(12, 16)$
13	14	$(8, 6) + (6, 8)$	$(8, 24) + (6, 32)$
17	18	$(12, 4) + (6, 8)$	$(12, 16) + (6, 32)$
19	20	$(12, 4) + (8, 6)$	$(12, 16) + (8, 24)$

TABLE 1. Class structure of the conic points of a zero divisor; stabilisers are $48/m$ and $192/m$ for $p \geq 5$.

mod 3) with $f(x\gamma) = \pi(\gamma)^{-1} f(x)$, in the conventions of [2]. For a zero divisor z with continuous stabiliser K_z, and $u \in \pi^{K_z}$, put

$$P_{z,u}(f) = \sum_{x \text{ compatible with } z} \langle f(x), u \rangle, \qquad x \text{ compatible} \iff \text{the line of } x \text{ lies in } \operatorname{Im} \mathbb{H}_z \bmod 3,$$

and more generally, for any orbit O of the finite group $\mathrm{Stab}(\mathbb{H}_z)$ on the structures, $P_{z,O}(f) = \mathrm{proj}_{\pi^{K_z}} \sum_{x \in O} f(x)$. Every zero divisor has exactly 4 compatible lines (the conic at $p = 3$).

5.2. **Trivial weight.** At the maximal parahoric level X_3 has two classes (stabilisers $432, 36$) and T_2-eigenvalues 126 (Eisenstein) and 9 (the lift of $3.6.a.a$). By Theorem 4.4 the four conic points of a Hurwitz zero divisor lie in the 432-class and those of a Lipschitz one in the 36-class, so $P_z(f) = 4f(\text{class of } z)$: for the cusp form, normalised $f = (-1, 1/12)$, the periods are -4 and $1/3$.

5.3. **Weights 14 and 27.** Weight 14 (adjoint): one form at level 3 ($T_2 = -18$, at the plane and flag levels), whose conic periods vanish for both types. Weight 27 ($2\omega_1$): at the Iwahori level a 7-dimensional space with $T_2 \in \{40.5\ (\times 2),\ 0\ (\times 3),\ -10.768,\ 6.268\}$, the irrational pair being the roots of $x^2 + 4.5x - 67.5$ and new at the Iwahori level; the others descend to the maximal parahoric levels, where $(T_2, T_5) = (81/2,\ 2034/5)$ and $(0,\ 11466/25)$ (twice), computed with the calibration $(9, 810)$ and $(126, 19530)$ in trivial weight.

Computation 5.1. (i) The conic period $P_{z,u}$ vanishes identically on the Iwahori-new forms and is nonzero on all old forms, for both types. (ii) Over all $\mathrm{Stab}(\mathbb{H}_z)$-orbits (51 orbits for Lipschitz, 21 for Hurwitz type), the Iwahori-new forms have nonzero $P_{z,O}$ on 41/51 resp. 16/21 orbits, vanishing exactly on the orbits containing the conic flags (and a few others); the old forms are nonzero on every orbit. (iii) For the Iwahori-new forms the sum of f over the four flags above *each* conic point vanishes separately, before any projection.

Remark 5.2 (what the conic vanishing is)**.** Item (iii) explains item (i) completely: the sum of f over the planes containing a fixed line is the degeneracy map from the Iwahori level to the maximal-parahoric line level, and Iwahori-new forms lie in its kernel for *every* line. The conic period factors through this degeneracy map, so its vanishing on new forms is the definition of newness, not a property of the zero divisors; the Hurwitz/Lipschitz type and the K-projection play no role in it. The content of zero-divisor periods is therefore (a) at the maximal parahoric level, where the conic law (Theorem 4.4) governs them, and (b) in orbit-periods $P_{z,O}$ that do not factor through degeneracy maps, which is where the new forms are nonzero, in accordance with the global expectation of the next subsection.

5.4. **The maximal parahoric level 7 and the selection rule.** At the maximal parahoric level $p = 7$ the class set has 8 classes (stabilisers $2, 18, 42, 6, 6, 2, 6, 24$), and the conics of the Hurwitz and Lipschitz zero divisors lie in the 24-class and in one 6-class (Theorem 4.4). Since the $p+1$ conic points form a single orbit of the finite group $\mathrm{Stab}(\mathbb{H}_z)$ when $p \le 11$, the conic sum $\sum_{\ell \in C_z} f(\ell)$ is the projection of $f(\ell_0)$ onto the $\mathrm{Stab}(\mathbb{H}_z)$-fixed vectors of V_π, which gives a selection rule: $\dim V_\pi^{\mathrm{Stab}(\mathbb{H}_z)}$ is 1 for the trivial weight, 0 for the 7 and the 14, and 1 (Hurwitz) or 2 (Lipschitz) for the 27. The rule for all weights up to 77, computed from the characters of $\mathrm{Stab}(\mathbb{H}_z)$ on the eigenvalues of its elements on the 7, is

weight	1	7	14	27	64	77	77′
$\dim V_\pi^{\mathrm{Stab}(\mathbb{H}_z)}$, Lipschitz ($\lvert\,\mathrm{Stab}\,\rvert = 48$)	1	0	0	2	1	1	3
$\dim V_\pi^{\mathrm{Stab}(\mathbb{H}_z)}$, Hurwitz ($\lvert\,\mathrm{Stab}\,\rvert = 192$)	1	0	0	1	0	0	2

so from weight 64 on the two types differ: in weights 64 and 77 every form is dark on Hurwitz conics and (generically) lit on Lipschitz ones. This was verified in weight 64 at level 7 (97 forms, the 64 cut out of $7 \otimes 14$ by the character projector over $G_2(\mathbb{Z})$): all 97 conic sums vanish for the Hurwitz type and none for the Lipschitz type. Accordingly all conic sums vanish identically in weights 7 and 14 for every form, and in weight 27 (50 forms at this level) the periods are nonzero for every form and both types; in trivial weight (7 tempered cusp forms) likewise. So zero-divisor periods do not vanish on tempered forms: they are generic nonzero functionals, in accordance with the nonvanishing predicted by the see-saw, and their content lies in the conic law and the selection rule rather than in any vanishing.

The same computation extends Theorem 5.3 to level 7: the five newforms of $S_{10}(\Gamma_0(7))$ appear in weight $2\omega_1$ at the maximal parahoric level with exactly the predicted (T_2, T_3), namely $(58.351, 116.103)$, $(37.021, 188.629)$, $(35.169, 28.970)$ (multiplicity 2), $(14.331, 164.808)$ (multiplicity 2) and $(1.378, 80.601)$; the first two violate the tempered bound, the others pass it. In trivial weight the level-7 lifts of weight-6 newforms have no fixed vector at this parahoric and live at the plane parahoric.

5.5. **Interpretation.** In [6] the pair $D^1 \subset \mathrm{Aut}(\mathbb{O})$ sits in a see-saw inside the adjoint E_7 of the Albert algebra, with $\mathrm{Aut}(\mathbb{O}) \times \mathrm{PGSp}_6$ and $D^1 \times G_D$ (G_D of type D_6) as dual pairs; the theta lift of the trivial representation of D^1 is a residue of a Siegel Eisenstein series on G_D, and by Pollack's integral and Chenevier's identity $L(s, \pi, \mathrm{Spin}) = \zeta(s) L(s, \pi', \mathrm{Std})$ for G_2-parameters, the D^1-period of a form on the compact G_2 is governed by $L(1, \pi', \mathrm{Std}) \neq 0$. Our periods are D^1-periods with specific local data at 3. Computation 5.1(ii) is in accordance: the genuine (Iwahori-new) forms have nonzero D^1-periods on orbits that do not factor through degeneracy maps; the vanishing along the conic is the degeneracy map itself (Remark above). The old forms are identified. Writing $\mathrm{SO}(4) = (\mathrm{SL}_2^{\mathrm{long}} \times \mathrm{SL}_2^{\mathrm{short}})/\mu_2 \subset G_2$ for the stabiliser of a quaternion subalgebra, with $7 = (2,2) \oplus (1,3)$ and $\mathrm{SL}_2^{\mathrm{long}} = D^1$, consider the Arthur-type parameter $\psi_f = \varphi_f \boxtimes \nu$ with a classical newform f of weight k on the long-root SL_2 and the Arthur SL_2 on the short root. Since the Hecke eigenvalue of an algebraic modular form is the Satake transform of the unitary local representation (the weight enters only at infinity), $T_p = p^3 \chi_7(s_p) - 1$ in every weight, and here

$$\chi_7(s_p) = \left(p + 1 + \tfrac{1}{p}\right) + \left(p^{1/2} + p^{-1/2}\right) \frac{a_p(f)}{p^{(k-1)/2}}, \qquad T_p = \frac{p^3(p^2+p+1)}{p} - 1 + \frac{(p+1)\,a_p(f)\,p^3}{p^{k/2}}.$$

Theorem 5.3. *At level* 3 *the trivial-weight cusp form* $(T_2, T_5, T_7) = (9, 810, 2472)$ *is* $\psi_{3.6.a.a}$ *$(a_2, a_5, a_7 = -6, 6, -40)$, and the weight-$2\omega_1$ old forms are* $\psi_{3.10.a.a}$ *$(a_2, a_5 = 18, -1530$; $T_2 = 81/2$, $T_5 = 2034/5)$ and* $\psi_{3.10.a.b}$ *$(a_2, a_5 = -36, -1314$; $T_2 = 0$, $T_5 = 11466/25$, multiplicity 2 at*

the maximal parahoric level), with the weight rule $k = 6 + 2a$ for G_2-weight $a\omega_1$. All six (resp. four) Hecke eigenvalues agree exactly; the prime 7 was not used in finding the dictionary.

These are the Gan–Gurevich non-tempered forms [5] seen on the compact G_2: the classical form lives on D^1, which is why their D^1-periods are large, and the two weight-10 forms have opposite Atkin–Lehner signs at 3 ($a_3 = \pm 81$), a natural source of the multiplicity difference (to be confirmed against Arthur's multiplicity formula). No level-one forms exist in weights 14, 27, 77′ ($G_2(\mathbb{Z})$-invariants vanish). The Iwahori-new irrational pair is not of this type and is a genuine (tempered) form.

6. The annihilator lattices, triality, and the next rung

Proposition 6.1 (annihilator lattices)**.** *Let $x = (a, b)$ be a zero divisor of the integral sedenions with unit halves and $\mathbb{H}_x = \langle 1, a, b, ab\rangle$ its quaternion subalgebra. The orthogonal complement $\mathbb{H}_x^{\perp} \cap \mathbb{O}_{\mathbb{Z}}$, whose units are exactly the partners c with $(c, \pm(ac)b)$ a zero divisor annihilating x, is the root lattice D_4 (determinant 4, 24 roots) when x is of Hurwitz type and the cubic lattice $\mathbb{Z}^4$ (8 units) when x is of Lipschitz type. Every imaginary unit lies in exactly 3 Hurwitz and 12 Lipschitz quaternion subalgebras, which gives the $12 + 48$ partners of a unit.*

Proposition 6.2 (the seven orbits, explained)**.** *For a Hurwitz zero divisor the 24 partners split as $8+8+8$ according to which of the three axes a, b, ab of the frame the partner c completes to a Hurwitz subalgebra; each partner completes exactly one axis, never two. These three classes are the three triality classes of the roots of D_4, and they are the three orbits of pairs with Hurwitz x. For a Lipschitz zero divisor the 8 partners are the $\pm$ unit vectors of $\mathbb{Z}^4$: three of the four axes carry the partners completing one frame axis each and the fourth carries the two partners completing all three, the pattern $(8, 24, 24, 24, 24)$. The missing eighth pattern $(24, 24, 24, 24, 24)$, an all-Hurwitz pair, cannot occur.*

Theorem 6.3 (four rungs of the Cayley–Dickson ladder)**.** *Let A_{2^k} ($k = 4, 5, 6, 7$) be the integral sedenions, 32-ions, 64-ions and 128-ions, the iterated Cayley–Dickson doubles of $\mathbb{O}_{\mathbb{Z}}$, with $m = 2^{k-3}$ octonion slots, and call an element $(x, y) \in A_{2^k} = A_{2^{k-1}} \oplus A_{2^{k-1}}$ unit-half if x and y are units of $A_{2^{k-1}}$ (the $240\,m/2$ elements with a root of E_8 in one slot). Then (x, y) is a zero divisor if and only if x and y are imaginary and perpendicular. The numbers of unit-half zero divisors are*

$$7{,}560, \qquad 98{,}280, \qquad 625{,}320, \qquad 3{,}061{,}800,$$

each equal to the number of ordered pairs of perpendicular imaginary units of the previous rung ($126 + 240(m/2 - 1)$ imaginary units; the pairs within the first slot are the 7,560 sedenion zero divisors, those within any other slot the 30,240 perpendicular pairs of E_8 roots, and any two different slots are perpendicular). The annihilator of a unit-half zero divisor has dimension $4 + 8t$ with $0 \le t \le m - 1$: $\{4\}$ at the sedenions, $\{4, 12\}$ at the 32-ions (59,472 and 38,808), $\{4, 12, 20, 28\}$ at the 64-ions (302,064, 171,360, 46,368, 105,528), and $\{4, 12, \ldots, 60\}$ at the 128-ions (1,077,936, 694,176, 290,304, 395,136, 76,608, 108,864, 172,704, 246,072), exactly the eight values $4 + 8t$ predicted before the computation. The strata are determined by the slots: two units in the same slot give the maximal dimension; two slots of one sedenion copy give 12; slots in different copies give 4, except that the 7,560 sedenion zero-divisor pairs placed in the two outermost slots give 20, and in the two innermost slots the pairs of distinct imaginary units give 20 ($126 \cdot 126 - 252$) and the coincident ones 28; the elements lying in a quaternion-sized subalgebra with a new unit ($b = \pm a$ or $b = \pm 1$) always reach the maximal dimension. In particular the annihilator of a sedenion zero divisor grows from 4 to 12 to 28 to 60 up the ladder. The automorphism group of the integral 2^k-ions, $G_2(\mathbb{Z}) \times (\mathbb{Z}/2)^{k-3}$, has 2, 24, 170 and 870 orbits on the unit-half zero divisors at $k = 4, 5, 6, 7$, that is $17m^2 - 29m + 14$ orbits with

$m = 2^{k-4}$ (a quadratic fitted on three rungs predicting the fourth), with stabilisers of orders $8, 32$*;* $6, 8, 16, 32, 192$*;* $12, 16, 32, 64, 384, 432$ *and* $24{,}192$ *(the six orbits of size* 4 *at the* 64*-ions are the real unit placed in two different slots).*

Proposition 6.4 (annihilator lattices up the ladder)**.** *The integral annihilator of a unit-half zero divisor, as a lattice of norm-*2 *vectors, is a root lattice, and the root systems are: at the sedenions* D_4 *(Hurwitz) and* A_1^4 *(Lipschitz); at the* 32*-ions* $D_4 \oplus E_8$ *and* $A_1^4 \oplus E_8$ *for the sedenion zero divisors, and* $D_6 \oplus D_6$ *for the perpendicular root pairs of the second copy and for the diagonal mixed elements; at the* 64*-ions* $D_4 \oplus E_8^3$ *for a same-slot Hurwitz element,* $D_6^2 \oplus D_4^2$ *for the sedenion pairs in the outermost slots and* $D_4^4 \oplus A_1^4$ *for the inner cross pairs of distinct imaginary units (the two* 20*-dimensional strata). The step of* 8 *in* $4 + 8t$ *is one copy of* E_8 *per octonion slot, and* D_6 *is the root system of* E_8 *perpendicular to a perpendicular pair of its roots.*

Relation to the literature. Over the reals the sedenion statement is classical: Moreno proved that zero divisors of the Cayley–Dickson algebras are imaginary with annihilators of dimension divisible by 4, and Biss, Dugger and Isaksen that (a, b) is a zero divisor exactly when a, b are imaginary, perpendicular and of equal norm, with $\dim \operatorname{ann} \leq 2^n - 4n + 4$ (G. Moreno, *The zero divisors of the Cayley–Dickson algebras over the real numbers*, Bol. Soc. Mat. Mex. 1998, and *Constructing zero divisors in the higher dimensional Cayley–Dickson algebras*, 2005; D. Biss, D. Dugger, D. Isaksen, *Large annihilators in Cayley–Dickson algebras*, Comm. Algebra 2008). Our dimensions $4, 12, 20, 28$ satisfy both constraints. What is new here is integral: the lattices D_4, A_1^4, their growth by E_8, the explanation of the seven orbits through the triality of D_4 and the axes of $\mathbb{Z}^4$, the exact unit-half census at the 32- and 64-ions, and the conic law and selection rule of the earlier sections.

Theorem 6.5 (structure of the annihilator, and the law at every rung)**.** *Let* x, y *be pure-slot units of* A_n *(an octonion unit in one slot of the standard basis),* $X = (x, y) \in A_{2n}$*, and* $Q = (z, w)$ *arbitrary. Then*

- *(i)* $N(XQ) = 2N(Q) - 2\langle z, (\bar{x}, \bar{w}, y)\rangle$*, where* $(a, b, c) = a(bc) - (ab)c$ *is the associator. Consequently* X *is a zero divisor if and only if some* $w \neq 0$ *is anti-associative with* $\bar{x}$ *and* y*, i.e.* $\bar{x}(\bar{w}y) + (\bar{x}\bar{w})y = 0$*, and then* $\operatorname{ann}(X) = \{(\bar{x}(\bar{w}y),\, w)\}$ *over the kernel of the anti-associator* $\Phi_{x,y} \colon w \mapsto \bar{x}(\bar{w}y) + (\bar{x}\bar{w})y$*.*
- *(ii)* $\Phi_{x,y}$ *maps each slot of* A_n *into a single slot (standard units multiply by symmetric difference of their index sets, up to sign), so* $\ker \Phi_{x,y}$ *is the direct sum of its single-slot kernels and* $\dim \operatorname{ann}(X) = \sum_R \dim \ker(\Phi_{x,y}|_{\text{slot } R})$*.*
- *(iii) Each single-slot restriction involves only* $\mathbb{O}$ *and at most three standard units, hence lives in a Cayley–Dickson subalgebra of dimension at most* 64*. The unit-half law and the stratification* $4 + 8t$ *therefore hold at every rung as soon as they hold for the* 64*-ions, which Theorem 6.3 establishes (the* 128*-ions serving as an independent check).*

Proof. We use the Cayley–Dickson product $(x, y)(z, w) = (xz - \bar{w}y,\ wx + y\bar{z})$, conjugation $\overline{(x, y)} = (\bar{x}, -y)$, the norm $N(x, y) = N(x) + N(y)$, and the adjoint identities $\langle ab, c\rangle = \langle b, \bar{a}c\rangle = \langle a, c\bar{b}\rangle$, valid in every Cayley–Dickson algebra (they follow by induction from the octonions, where they are standard).

Step 1: pure-slot elements multiply norms on both sides. Call $u \in A_n$ pure-slot if it lies in one octonion slot of the standard basis. We show $N(uq) = N(u)N(q) = N(qu)$ for all q, by induction on n; for $n = 3$ this is the composition property of the octonions. A pure-slot element of A_{2n} is $(u', 0)$ or $(0, u')$ with u' pure-slot in A_n, and for $q = (z, w)$:

$$(u', 0)(z, w) = (u'z,\ wu'), \quad (0, u')(z, w) = (-\bar{w}u',\ u'\bar{z}), \quad (z, w)(u', 0) = (zu',\ w\bar{u}'), \quad (z, w)(0, u') = (-\bar{u}'w,\ u'z).$$

Each component is u' times (or times u') an element of A_n of norm $N(z)$ or $N(w)$, so by induction every right-hand side has norm $N(u')(N(z)+N(w))$.

Step 2: the identity. For $X=(x,y)$ with x,y pure-slot units and $Q=(z,w)$,

$$N(XQ)=N(xz-\bar{w}y)+N(wx+y\bar{z})=N(xz)+N(\bar{w}y)-2\langle xz,\bar{w}y\rangle+N(wx)+N(y\bar{z})+2\langle wx,y\bar{z}\rangle,$$

and by Step 1 the four norms are $N(z),N(w),N(w),N(z)$. By the adjoint identities, $\langle xz,\bar{w}y\rangle=\langle z,\bar{x}(\bar{w}y)\rangle$ and $\langle wx,y\bar{z}\rangle=\langle\bar{z},\bar{y}(wx)\rangle=\langle z,\overline{\bar{y}(wx)}\rangle=\langle z,(\bar{x}\bar{w})y\rangle$, so

$$N(XQ)=2N(Q)-2\langle z,\ \bar{x}(\bar{w}y)-(\bar{x}\bar{w})y\rangle=2N(Q)-2\langle z,(\bar{x},\bar{w},y)\rangle.$$

Step 3: the criterion. By Step 1, $N(\bar{x}(\bar{w}y))=N((\bar{x}\bar{w})y)=N(w)$, so $N((\bar{x},\bar{w},y))\le 4N(w)$, and by Cauchy–Schwarz $\langle z,(\bar{x},\bar{w},y)\rangle\le 2\sqrt{N(z)N(w)}\le N(z)+N(w)=N(Q)$. Hence $N(XQ)\ge 0$ with equality if and only if $N(z)=N(w)$, $(\bar{x},\bar{w},y)$ has norm $4N(w)$, and z is a positive multiple of it; the middle condition says $\bar{x}(\bar{w}y)=-(\bar{x}\bar{w})y$, i.e. $\Phi_{x,y}(w)=0$, and then $(\bar{x},\bar{w},y)=2\bar{x}(\bar{w}y)$ and $z=\bar{x}(\bar{w}y)$. This proves (i).

Step 4: slots. The standard basis of A_n consists of the products $u\,e_S$ of octonion basis elements with the standard units e_S indexed by subsets S of the doubling generators, and $e_Se_T=\pm e_{S\triangle T}$. If $x=ae_S$, $y=be_T$ and w lies in slot R, both $\bar{x}(\bar{w}y)$ and $(\bar{x}\bar{w})y$ lie in slot $S\triangle R\triangle T$, and distinct R give distinct target slots. So $\Phi_{x,y}$ is a direct sum of maps between slots and its kernel is the direct sum of the single-slot kernels, which is (ii). Finally the restriction of $\Phi_{x,y}$ to slot R is computed inside the subalgebra generated by $\mathbb{O}$ and e_S,e_T,e_R, a Cayley–Dickson algebra of dimension at most 64, which is (iii). □

Remark 6.6. By Theorem 6.5, the unit-half law and the stratification $4+8t$ hold at every rung of the integral Cayley–Dickson ladder, modulo the standard facts about the standard basis used in (ii) and (iii); the stratum t counts the slots R whose anti-associator block is singular.

The law thus persists through the rung (32-ions) where the norm-counting Euler product is lost ($E_4^4=E_{16}+\frac{3456000}{3617}E_4\Delta$): multiplicativity and the zero-divisor geometry are independent phenomena.

Computations: `zd32.py` and `zd64.py` in the repository (left-multiplication matrices of the 32- and 64-ions on the 230,400 and 921,600 unit-half candidates; annihilator dimensions by rank; the law verified anchor by anchor at all three rungs).

7. Negatives and boundaries

7.1. **Sedenion spectral geometries.** Farnsworth and Boyle [8, 9] extend Connes' spectral triples to non-associative algebras: the Hilbert space must be a bimodule in the algebra's own class (every identity surviving the replacement of one algebra element by a vector), the order conditions become vanishing associators in $B=\Omega A\oplus H$, and the gauge group is $\mathrm{Aut}(A)$. For the sedenions acting on themselves: the flexible order-zero law holds and the alternative ones fail; the derivation algebra is 14-dimensional and inner (inside $\mathfrak{so}(16)\oplus\mathbb{R}$); the fermions decompose as $2(1+7)$ under G_2 with the S_3 of $\mathrm{Aut}(\mathbb{S})=G_2\times S_3$ as a family symmetry; the commutant of the left action is trivial, forcing $\gamma=\pm1$ and $D_F=0$. On two copies $\mathbb{S}\oplus\mathbb{S}$ with $\gamma=\mathrm{diag}(1,-1)$, the flexible order-one condition forces D_F to be a multiple of the identity for either real structure; with the opposite bimodule on the second copy, no D_F survives; on the complex sedenions $\mathbb{C}\otimes\mathbb{S}$ with real-linear D_F, only complex scalars survive. The sedenion analogue $H_3(\mathbb{S})$ of the Albert algebra violates the Jordan identity (relative size 0.96; $H_3(\mathbb{O})$ satisfies it to 10^{-15}), so the Jordan-geometry route [9] excludes the sedenions and admits the Albert algebra. In every construction the zero divisors do not enter the action.

7.2. **Euler products along the ladder.** The norm-counting Dirichlet series $\sum r(n)n^{-s}$ has an Euler product iff r is multiplicative. It does for $\mathbb{Z}$, $\mathbb{Z}[i]$, the Hurwitz quaternions (D_4), the Coxeter octonions (E_8, $r = 240\sigma_3$) and the integral sedenions ($E_8 \oplus E_8$, $r = 480\sigma_7$) — the last only because $M_8(\mathrm{SL}_2(\mathbb{Z}))$ has no cusp forms — and fails for the integral 32-ions (E_8^4), since $E_4^4 = E_{16} + \frac{3456000}{3617}\,E_4\Delta$. The counting zeta function of the 32-ions, $\frac{16320}{3617}\zeta(s)\zeta(s-15) + c\,L(E_4\Delta, s)$, has two zeros on its critical line $\Re s = 8$ below $t \approx 15$ and none up to $t = 100$, where the Eisenstein part dominates by 10^5–10^6; the Leech lattice's $\zeta(s)\zeta(s-11) - L(\Delta, s)$ has no zeros at all on $\Re s = 6$ to $t = 120$, its 57 zeros to height 198 lying near $\Re s \in [11.1, 12.5]$ with spacings more rigid than GUE. The product over primes is lost exactly where Δ enters.

8. Open problems

(1) A conceptual proof of Lemma 4.5 replacing the finite computation at $p = 5, 7$ (the primes 5 and 7 divide some saturation indices without producing isotropic intersections).
(2) The multiplicities of the Arthur-type lifts of Theorem 5.3 against Arthur's multiplicity formula ($a_3 = \pm 81$), and the identification of the Iwahori-new tempered pair through its exceptional theta lift to genus-3 Siegel forms of level 3.
(3) The 32-ions; the Okubo integral orders of [12] and their idempotent geometry against the conic law.
(4) Part III: the J_E half of the F_4 class set at level 5 (88% of the mass found; exact deficits 720799/96768 and 3).

Appendix A. Code and data

Repository `chrisyounge-create/Cayley-Agent-Repository`: `sedenion.py` (doubling, laws), `sedenion_geometry.py`, `sedenion_step2.py`, `sedenion_step3.py`, `sedenion_step3_complex.py` (spectral geometries), `zd_hecke.py` (stabiliser, annihilator, multiplicities), `g2table/zd_periods.py`, `g2table/zd_periods_vec.py` (periods), `g2table/lemma43.py` (proof of Lemma 4.5), `g2table/zd_periods_level` (level-7 periods and lifts), `beurling.py` (prime-correlation model). Dropbox notes `RH/SEDENIONS.md` through `RH/NEXT_RUN_ITEMS_1_TO_4.md` contain the session records.

References

[1] C. Younge, *Class sets and Eisenstein congruences on compact exceptional groups*, Part I of this record (2026).
[2] C. Younge, *The Hecke module of the compact F_4 at level 3*, Part II of this record (2026).
[3] G. Moreno, The zero divisors of the Cayley–Dickson algebras over the real numbers, Bol. Soc. Mat. Mexicana 4 (1998).
[4] R. B. Brown, On generalized Cayley–Dickson algebras, Pacific J. Math. 20 (1967).
[5] W. T. Gan, N. Gurevich, Non-tempered Arthur packets of G_2: liftings from $\widetilde{\mathrm{SL}}_2$, Amer. J. Math. 128 (2006).
[6] W. T. Gan, G. Savin, An exceptional Siegel–Weil formula and poles of the spin L-function of PGSp_6.
[7] G. Chenevier, Subgroups of Spin(7) or SO(7) with each element conjugate to some element of G_2, and applications to automorphic forms, Doc. Math. (2019).
[8] S. Farnsworth, L. Boyle, Non-associative geometry and the spectral action principle, JHEP 07 (2015) 023, arXiv:1303.1782.
[9] L. Boyle, S. Farnsworth, Non-commutative geometry, non-associative geometry and the standard model of particle physics, New J. Phys. 16 (2014) 123027.
[10] N. Gillard, N. Gresnigt, Three fermion generations with two unbroken gauge symmetries from the complex sedenions, Eur. Phys. J. C 79 (2019) 446.
[11] C. Furey, Three generations, two unbroken gauge symmetries, and one eight-dimensional algebra, Phys. Lett. B 785 (2018).
[12] D. Corradetti, A. Marrani, Integral orders for the Okubo algebra and idempotent geometries of the E_8 lattice, arXiv:2608.27762 (2026).
[13] J. H. Conway, D. A. Smith, *On Quaternions and Octonions*, A K Peters (2003).

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